Roll No. ....

TCE–502

B. TECH. (CIVIL ENGINEERING) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

REINFORCED CEMENT CONCRETE—I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

(iii) IS 456 : 2000 issued by dept. is allowed for use.

1. (a) State and explain any *five* conventional, five advanced NDT testing methods in deciding the concrete quality. 10 Marks (CO1)

OR

(b) Decide the ingredient proportions by weight, for M25 grade concrete with the given data : 10 Marks (CO1)

Water-Cement Ratio 0.5	Cement	Fine Aggregate	Coarse Aggregate
Void Ratio	45%	50%	55%
Density (kg/m^3)	1800	1600	1700

2. (a) Define and explain modulus of elasticity, and how is it concerned for use in RCC design ? 10 Marks (CO1)

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OR

(b) Determine the standard cube characteristic strength for the given observations of samples : 10 Marks (CO1)

16.7	20.4	18	27.4	17.2.3	22	14.7	17.4
18.3	14.9	13.2	25.8	19.3	21.4	18.6	14.8
21.5	16.4	19.9	22.3	18.6	20.4	16.2	18.1
24.8	20.1	22	24.8	12.2	22.7	18.5	19.4
16.8	21.6	17.9	27.2	15.8	29.4	17.6	17.6

3. (a) Illustrate the concept of balanced, under reinforced and over reinforced section for flexure, with a brief discussion of the failure prospects for design. 10 Marks (CO1)

OR

(b) Formulate the stress block parameters for flexure in limit state method of design. 10 Marks (CO2)

4. (a) Design for flexural reinforcement at the mid-span section of a singly reinforced rectangular beam of 6 m effective span and is subjected to a dead load of 3 kN/m and a central concentrated load of 50 kN in working stress method of design. Use M20 and Fe415 grade materials.

10 Marks (CO2)

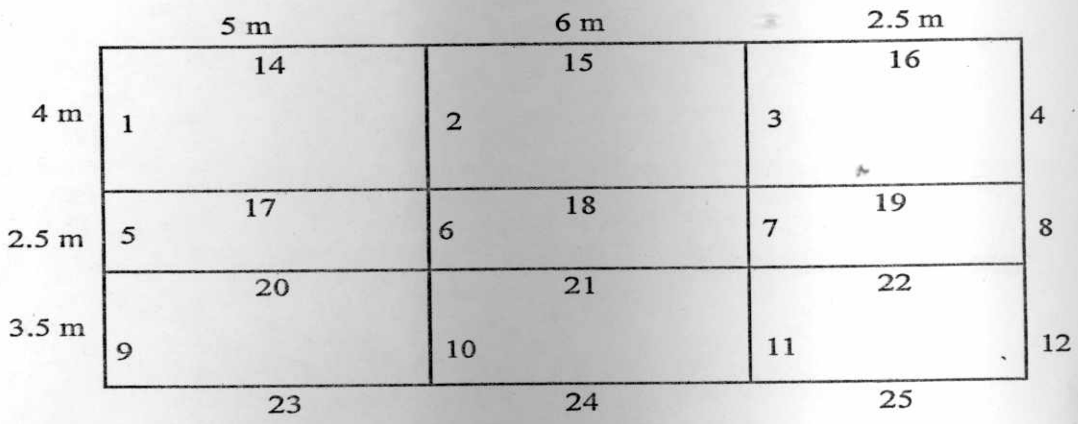
OR

(b) Decide the moment carrying capacity of the given rectangular beam section in LSM method of design.

Effective Span = 5 m, Live Load = 8 kN/m run, Cross section 250 × 400 mm with a clear bottom cover of 25 mm and 3 No's of 20 mm dia bars are used as main reinforcement in the section. M25 and Fe415 grade materials are used. 10 Marks (CO2)

5. (a) Design beam no. 18 shown in the plan below, as a singly reinforced rectangular beam section in LSM to support the slab, with the following loading conditions :

Live load on the slab is 2.5 kN/m^2, Floor Finish Load is 1 kN/m^2, Slab thickness is 120 mm casted in M20 grade concrete and Fe415 grade steel. 10 Marks (CO2)



OR

(b) Write a short note on detailing recommendations of IS code for main, shear reinforcement in Flexural members. Draw the typical reinforcement arrangement for a simply supported and cantilever beam.

10 Marks (CO2)

90